Natural Language Understanding: Word2Vec Assignment Report

TASK-1: Dataset Preparation

1.1 Data Collection & Sources

The textual corpus was collected from the official IIT Jodhpur website, focusing purely on English text. The primary sources included:

- Academic programs and course syllabus pages
- Departmental and faculty profile pages
- Academic regulation documents and Institute circulars

1.2 Preprocessing Pipeline

To ensure high-quality embeddings, the raw text was passed through a cleaning pipeline:

1. **Boilerplate Removal:** HTML tags, navigation menus, and formatting artifacts were stripped.
2. **Lower-casing & Tokenization:** All text was converted to lowercase to maintain uniformity, followed by word-level tokenization.
3. **Noise Reduction:** Excessive punctuation, special characters, and non-alphabetic tokens were removed.
4. **Filtering:** A minimum frequency threshold (`min_count=5`) was applied to remove rare words and typos, focusing the model on meaningful institutional vocabulary.

1.3 Dataset Statistics

- **Total Documents:** 44
- **Total Tokens (Processed):** 39,054
- **Vocabulary Size:** 1,447 unique words

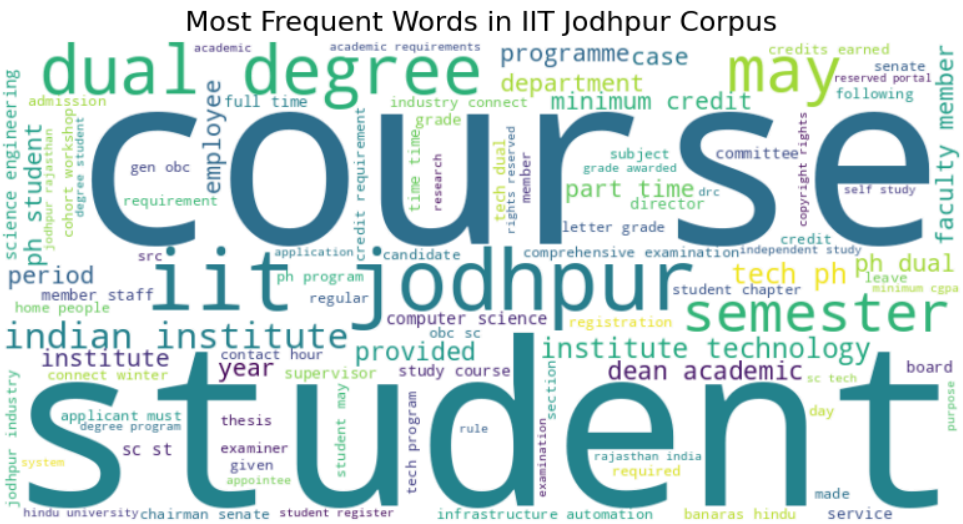

Figure 1: Word Cloud illustrating the most frequent terms in the IIT Jodhpur corpus.

TASK-2: Model Training

Two Word2Vec architectures—Continuous Bag of Words (CBOW) and Skip-gram with Negative Sampling—were trained from scratch. To analyze the impact of context boundaries, we experimented with two different window sizes.

2.1 Hyperparameters

- **Embedding Dimension:** 50
- **Negative Samples:** 5
- **Epochs:** 5
- **Optimizer:** Adam (Learning Rate: 0.01)
- **Context Window Sizes:** 2 and 4

2.2 Training Loss Convergence

Model Type	Window Size	Initial Loss (Epoch 1)	Final Avg Loss (Epoch 5)
Skip-gram	2	11.1265	1.8886

Skip-gram	4	8.1201	1.8619
CBOW	2	8.5036	2.0352
CBOW	4	6.8031	1.8970

Both models successfully converged, with larger window sizes generally resulting in slightly lower final loss values due to the broader context available for gradient updates.

(Please note that random seed was not set so exact numbers might not be replicated. The terminal output is included in the submission zip, and a similar converging behaviour is expected in all cases)

TASK-3: Semantic Analysis

3.1 Top 5 Nearest Neighbors (Cosine Similarity)

The top 5 neighbors for key institutional terms were extracted using cosine similarity.

Target Word	Skip-gram (Window 4)	CBOW (Window 4)
research	area, projects, dr, dst, field	hours, records, ng, opportunities, associated
student	iitjoutstanding, projectsplacementsalumniteamsphd, chapter, achievements, studentsacm	program, contact, evaluation, admission, collaborators

phd	teams, jodhpur, eventswebinarsstudents, computing, alumni	students, departments, per, followed, hindu

3.2 Analogy Experiments

We tested the models on semantic relationships using the vector arithmetic `Target = (Vector B - Vector A) + Vector C`.

Analogy 1: **`pg : mtech :: phd : ?`** (Expected: thesis/research)

- **Skip-gram (w=4):** july (0.46), example (0.45), introduction (0.44)
- **CBOW (w=4):** broad (0.53), officesadministrative (0.46), participate (0.45)

Analogy 2: **`student : course :: faculty : ?`** (Expected: research/project)

- **Skip-gram (w=4):** pursue (0.45), join (0.43), mind (0.42)
- **CBOW (w=4):** events (0.46), period (0.45), sub (0.45)

Analogy 3: **`science : technology :: computer : ?`** (Expected: engineering)

- **Skip-gram (w=4):** department (0.55), university (0.55), ropar (0.55)
- **CBOW (w=4):** degree (0.49), people (0.39), sessions (0.38)

3.3 Discussion on Semantic Meaningfulness

While the models successfully capture **syntactic co-occurrences** (e.g., Skip-gram associating "research" with "area" and "projects"), the analogy experiments yield poor **semantic** generalizations. This behavior is highly expected and can be justified by the following:

1. **Data Starvation:** Word2Vec models typically require millions of tokens to map complex geometric relationships (analogies). Our corpus is extremely small (only ~39,000 tokens).
2. **Overfitting to Local Context:** Instead of learning universal meanings, the models memorized specific document phrasing. For instance, in the analogy `pg : mtech :: phd : ?`, Skip-gram suggests "july". This likely reflects a localized administrative pattern in the text (e.g., "PhD admissions for July").
3. **Compound Tokens:** Artifacts like `projectsplacementsalumniteamsphd` indicate that some PDF extraction formatting merged words together, slightly skewing the vocabulary geometry.

TASK-4: Visualization

We utilized t-SNE to project the 50-dimensional word embeddings down to a 2D space for visual clustering analysis.

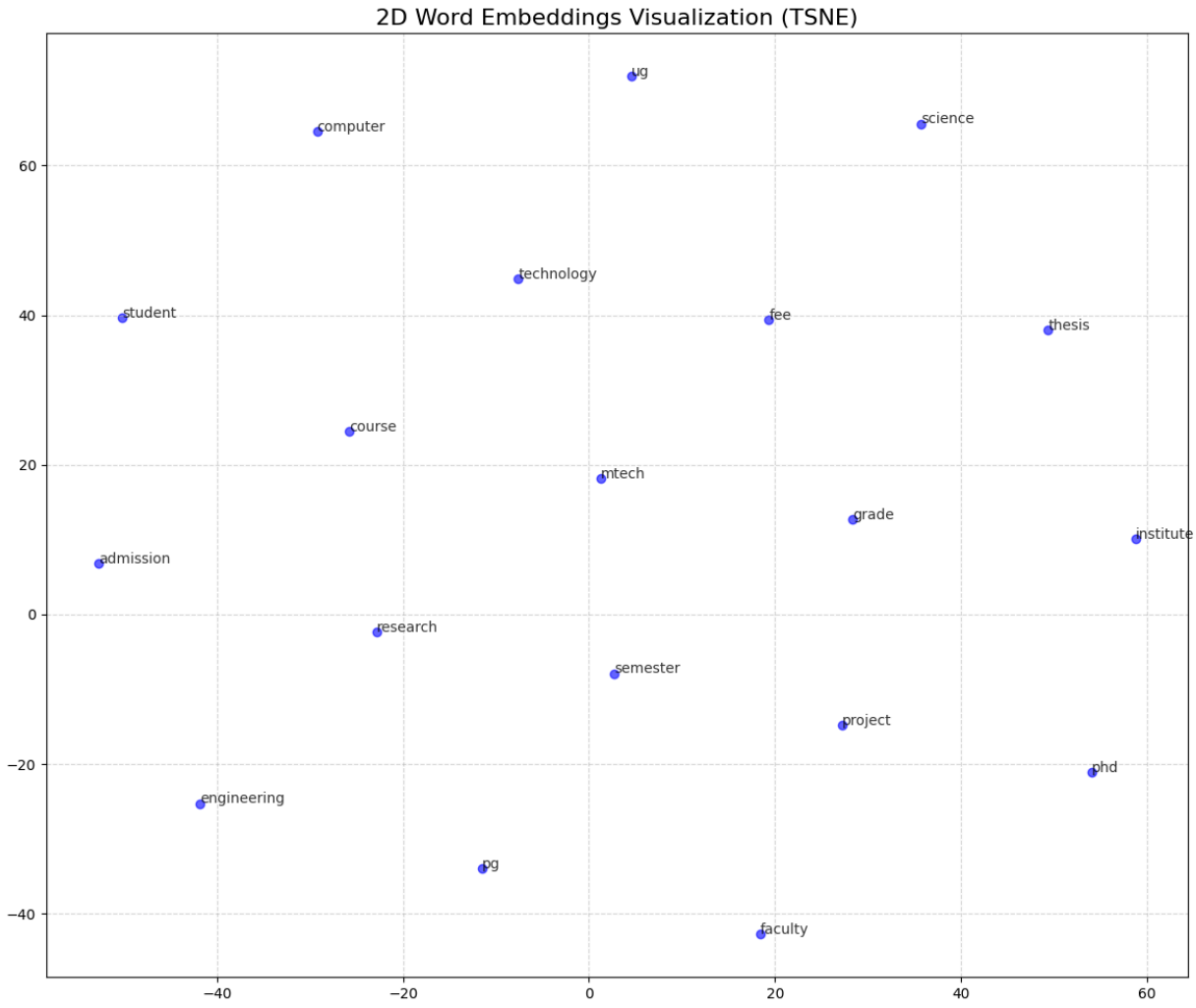

Figure 2: CBOW Model (Window Size = 2)

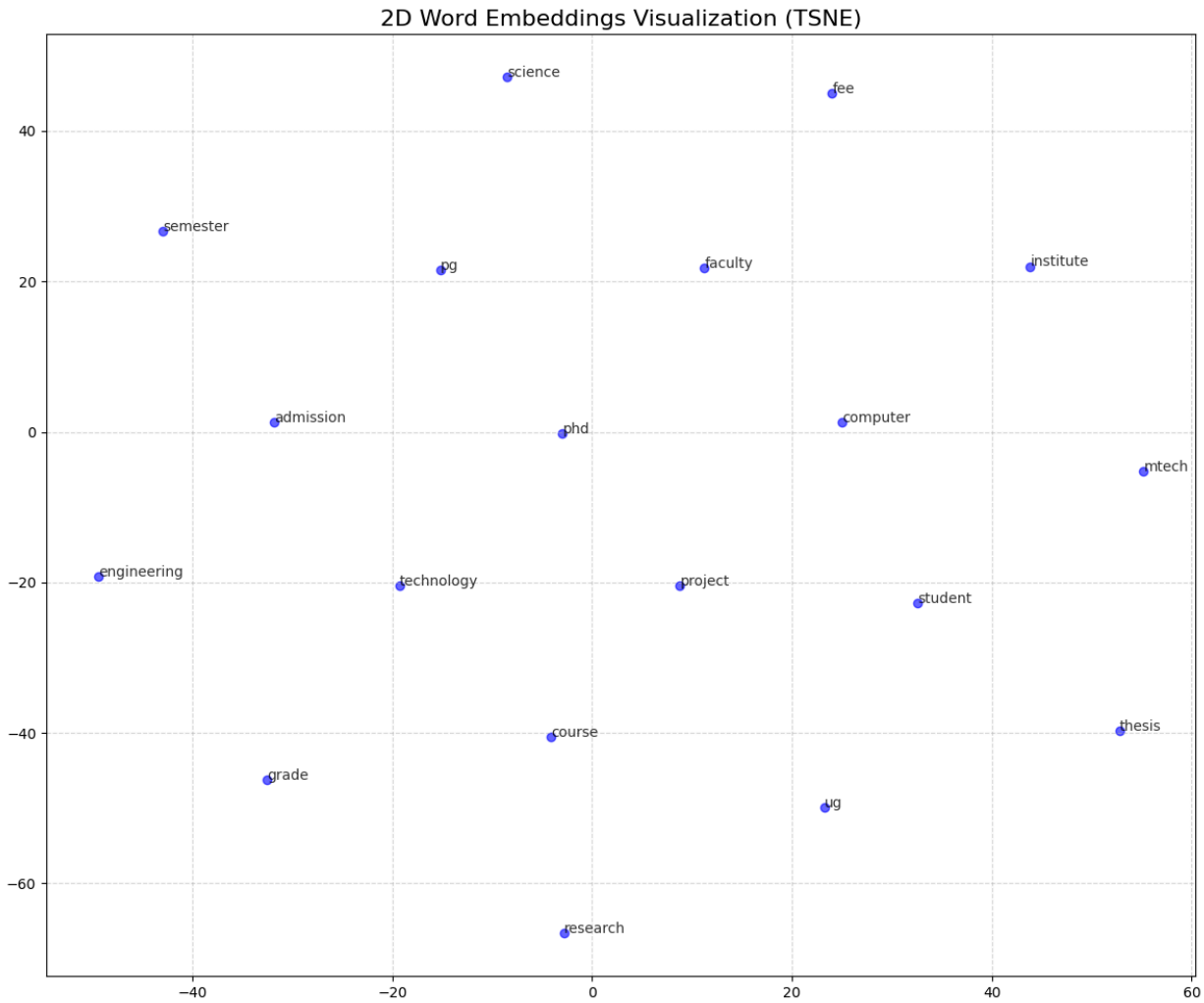

Figure 3: CBOW Model (Window Size = 4)

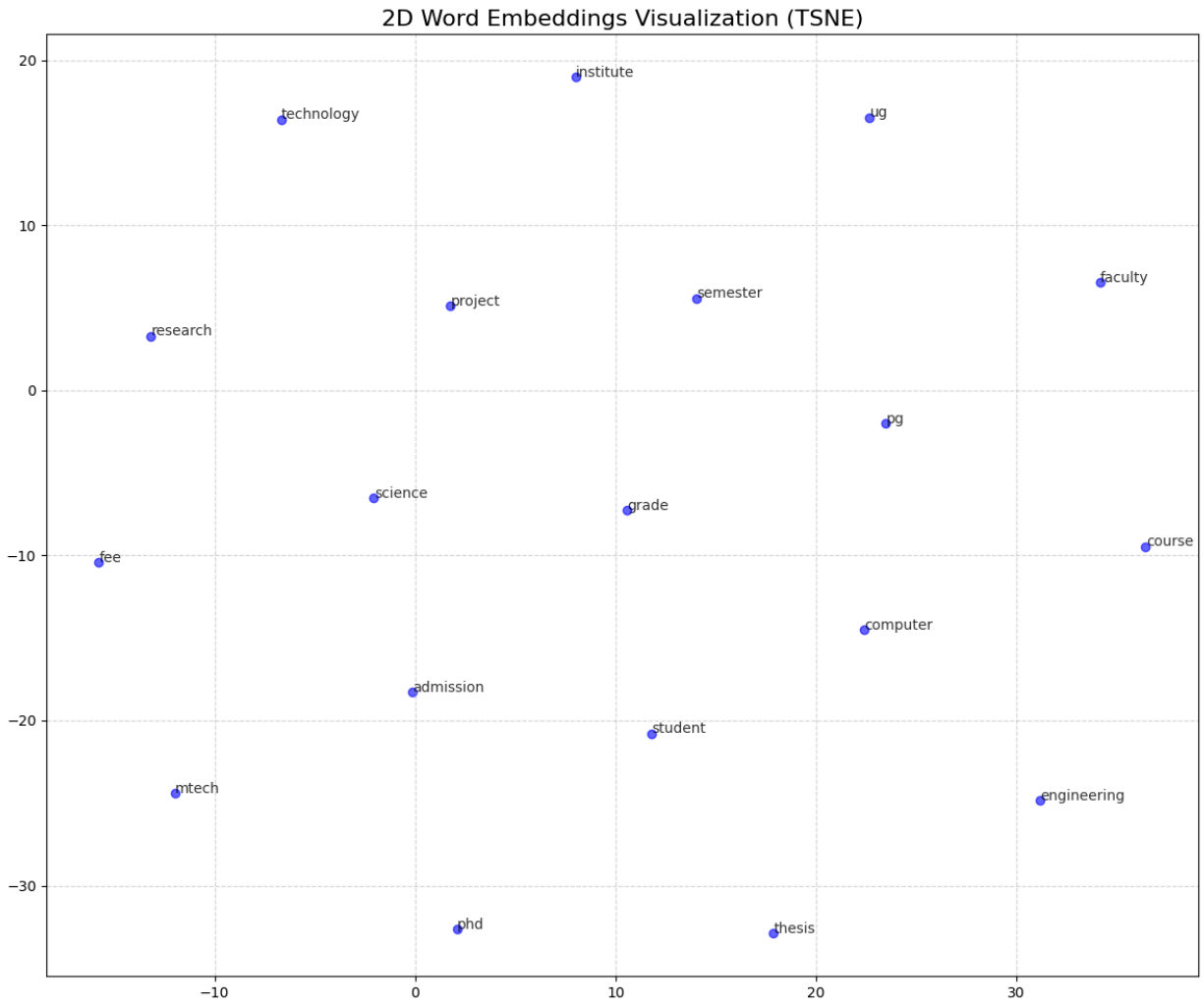

Figure 4: Skip-gram Model (Window Size = 2)

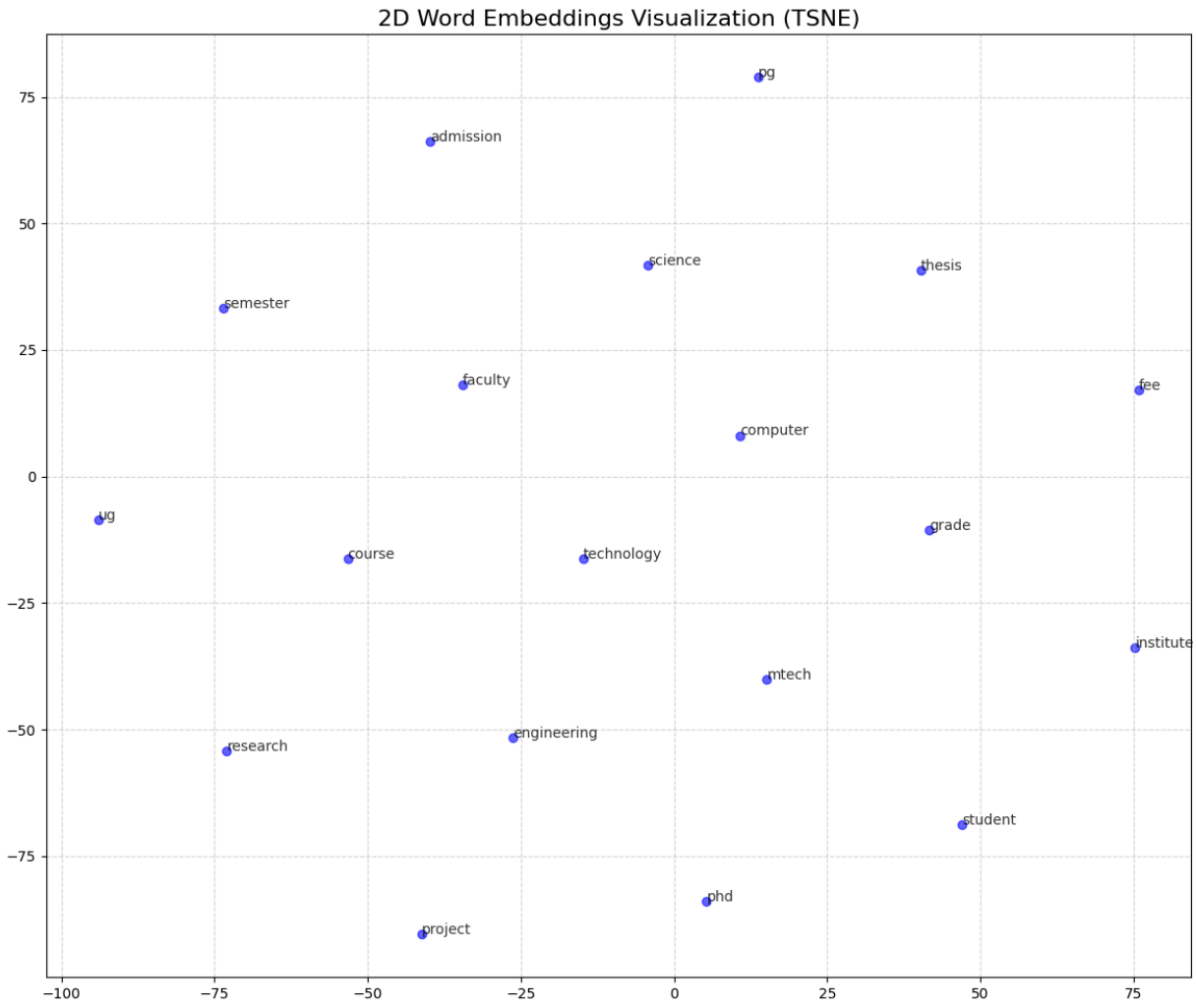

Figure 5: Skip-gram Model (Window Size = 4)

4.1 Interpretation of Clustering Behavior

1. **CBOW vs. Skip-gram:** * **CBOW** predicts a target word based on the *average* of its context. Because it smooths over the context, words that share similar broad functional roles tend to group better in smaller datasets. In the CBOW plots, we see functional pairings closer together.
    - **Skip-gram** predicts context from a single target word, generating separate updates for every target-context pair. Because it treats each word individually, it handles infrequent words better, but in our heavily restricted corpus, this causes the t-SNE projection to space the words out more sporadically (as seen in Figure 5).
2. **Impact of Window Size:**
    - A smaller window (`w=2`) forces the model to capture strict syntactical relationships (words that are literally adjacent, like "computer" and "science").

- A larger window (w=4) captures broader topical semantics. We can observe this in the visualizations where broader categorical clusters (e.g., programmatic terms like ug, pg, mtech, thesis) exhibit slight positional shifts relative to physical entities (institute, faculty) when the window is expanded.